

# Mitigating Bias in Technical Recruitment through a Unified, Dynamic Assessment Platform

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**Abstract**—In today’s modern recruitment apps, job postings, CV filtering, assessment tests and interview reviews spread over disjointed processes. This leads to slow hiring, poor elements of fairness, lack of transparency and inconsistent decision-making. We propose a system to integrate mediation across the entire recruitment cycle, starting from recruiter initiated job details and candidate job applications, to structured assessments and controlled handoffs.

The system offers a recruiter and a candidate portal, role-based access control and an adjustable job definition module. Evaluation of candidates is carried out through a multi-staged process that includes resume filtering via skill matching, a funnel of assessment tests, a brief structured interview and a combined score. The evaluative criteria engine is rule-based to ensure screening is the same for all candidates and explainable. Gap filling during assessment is based on a machine learning and NLP hybrid to refine and rank within a set of qualifying candidates without violating hard constraints. Edged case tests and defensive input management support the claims of system robustness. The entire platform is in line with modern recruitment ethics, scalable and offers a high degree of automation while preserving the needed control and transparency from the recruiter.

**Index Terms**—Recruitment automation, rule-based evaluation, weak supervision, hybrid ML and NLP ranking, candidate assessment and explainable systems

## I. INTRODUCTION

Ongoing increases in applicant volume, variety in job descriptions and expedited hiring needs have made recruitment workflows more challenging. Most current recruitment tools address this challenge by automating resume gathering and initial screening while leaving more advanced assessments and interviews to be conducted using other tools. This approach increases fragmentation, decreases evaluative consistency across multiple stages, decision latency and increases recruiter process heaviness.

In most cases, hiring decision makers in preliminary stages get influenced by formatting, keywording and prestige bias in applicant resumes without utilizing validated competencies.

Candidates also frequently complain about inadequate feedback, lack of status updates and evaluation transparency. These pain points present an opportunity for recruitment tools that have structured workflows, objective screening and decision auditability.

The incorporation of intelligent elements in recruitment systems is driven by the latest developments in automation and data-centric evaluation tools. However, the use of black-box or fully automated systems raises issues regarding ethical fairness, explainability and loss of control from the recruiter. There is, therefore, an increasing interest in recruitment systems where automation is supplemented with human control, configurable rules and logic.

Given these insights, we design and propose an integrated recruitment mediation system. This system is intended to assist in managing all aspects of the recruitment lifecycle within a single solution. This system is a recruiter-managed configuration of the job with candidate job applications, evaluation and seamless gating to facilitate tiered processing and unified ranking. The initial evaluation stages employ deterministic rule-based screening to ensure transparency and fairness, while selectively using machine learning and natural language processing to enhance candidate ranking within the confines of hard eligibility requirements. The proposed system seeks to boost efficiency, reliability and consistency in recruitment by blending structured decision-making with controlled intelligent systems.

## II. LITERATURE REVIEW

### A. Recruitment Automation and Traditional Hiring Systems

The increase in the number of job applications and the inefficiency in manual screening processes has drawn more attention to recruitment automation. With the traditional recruitment platform, the primary function is to collect resumes and the recruiters have to manually filter candidates on the basis of keywords, level of education and previous experience. This method is subjective, inconsistent and spillage of time.

Studies have shown that early stage hiring decisions are frequently influenced by non-skill related factors which reduces fairness and transparency in candidate evaluation [1] [2].

The Applicant Tracking System (ATS) was created to help simplify the recruitment process by offering a central place to track applications and store resumes. However most ATS platforms focus on operational efficiency rather than evaluation intelligence. They lack integrated mechanisms for skill validation unified scoring or structured decision making across multiple recruitment stages [14]. As a result recruiters still rely on fragmented tools such as external assessments informal interviews and manual judgment which leads to incomplete candidate profiles and delayed hiring outcomes.

These restrictions indicate that there is a need for recruitment frameworks that surpass single resume collection and equipped with evaluative workflows that allow for objective screening and consistent decision making.

### *B. Resume Parsing and Skill Extraction Techniques*

Resume screening is one of the main processes used by automated recruitment tools. The first systems encountered problems in resume scanning due to keyword matching and ruled based systems. These systems did not take into consideration variations in resume layouts and styles of writing. Recently, a number of studies have focused on different NLP and machine learning techniques to identify and categorize skills and other relevant data from unstructured text in resumes [3] [7].

Transformers and other deep learning based models have shown to be very successful in named entity recognition and other relevant fields in resumes [6]. The benefits of these systems are improved recall and understanding of the resume. However, these benefits are oftentimes associated with a lack of transparency and a greater need for computational power and labeled data. These factors, especially in the context of high-stakes recruitment are a source of concern and can undermine trust in real-life recruitment scenarios.

In addition to these models, others have tried to find the relationships between skills, positions and the level of experience of users in their professional networks [8]. These models, while improving the context of their understanding, are oftentimes black-box processes and can make it challenging to understand the reasons for their decisions.

Current research omits important insights in resume intelligence by treating parsing as a stand alone process, ignoring the entire recruitment decision process.

### *C. Automated Candidate Screening and Scoring Systems*

Several studies have proposed automated resume screening and candidate ranking systems using machine learning models and similarity metrics [13] [17]. These systems typically generate relevance scores between resumes and job descriptions and use these scores to rank candidates. While such approaches improve efficiency they frequently rely on single stage evaluation which limits robustness and increases sensitivity to noisy inputs.

Automation of candidate scoring has shown a need for objective and consistent scoring across evaluations [5]. Systems that score using a single metric might place too much emphasis on some signals and completely miss others resulting in scoring that is unfair. Although multi factor evaluation systems have been proposed to address these issues, their lack of clarity, logic and transparency in weight assignment are major issues.

These insights help in understanding the need for a structured scoring system that incorporates various evaluation dimensions and combines them with recruiter discretion and scorability.

### *D. Technical Assessments and Coding Evaluation*

The purpose of technical assessments is to see if a candidate has the knowledge to support their resume. Some automated coding tests have been suggested to evaluate and grade programming skills without the issues of cheating or program execution [11]. Some of the more recent literature suggests that coding assessments should evaluate more than just correctness; they should evaluate how the code performs and how the code is structured [12].

The literature suggests that recruitment systems have the technical assessments done outside the system or uniformly for a given role without regard to the function. This creates a lack of relevant assessments and increases the frustrating burden on candidates. This technique is in opposition to how real world assessments work. Literature suggests that the system should be able to configure the technical assessments and more importantly, the assessment should take context into account to be more in line with the current state of hiring.

### *E. Interview Automation and Behavioral Evaluation*

Automated interview systems have explored the use of speech facial expressions and behavioral cues to predict candidate suitability [9]. While such approaches aim to standardize interview evaluation they introduce significant concerns related to cultural bias demographic sensitivity and ethical compliance. Studies evaluating video based interview assessment systems report risks of biased predictions and limited generalizability [10].

As a result several researchers argue that interview automation should be used cautiously and should not replace human judgment in hiring decisions. Using structured interview frameworks with fixed prompts and minimal automation is suggested as a safer option.

### *F. Bias Fairness and Ethical Considerations in Recruitment Systems*

In AI driven recruitment, bias and fairness are still challenges. Studies show that if the logic of the decision process isn't designed well, hiring automation will increase the bias that already exists in social and organizational structures [1] [2]. There are some suggestions, such as the use of fairness metric and multi criteria decision making, which have been proposed to lessen the bias [4] [16].

In recruitment automation, the ethical assessments that are carried out, show the need for Human supervision, along with, clarity and accountability. Fully automated systems tend to lack these traits, along with the capability of being compliant to regulation. This evidence advocates for blended systems that pair automation with clear, recruiter-set parameters and rules.

#### G. Hybrid Machine Learning Approaches in Recruitment

Recent studies examine combined rule based logic and machine learning augmented hybrid recruitment systems [15] [18]. These systems use ML models to refine ranking or similarity estimation while preserving hard constraints enforced by rules, attempting to strike a balance between understanding the semantics and the interpretability of the ML models.

To address this problem, utilizing weak supervision from such deterministic systems has been suggested to decrease the need for hand-annotation and increase the level of scalability [16]. In recruitment, where labeled data is limited and often bias, these methods are especially useful.

The best hybrid systems rankings while keeping the system transparency and control that are increasingly demanded from recruitment automation systems.

#### H. Summary and Research Gap

Existing literature provides extensive coverage of individual recruitment components such as resume parsing automated screening technical assessments and interview analysis. However most systems address these components in isolation and lack an integrated end-to-end recruitment workflow with unified scoring and controlled decision making.

There remains a clear research gap for recruitment platforms that combine structured multi stage evaluation deterministic eligibility logic configurable assessments and hybrid ML-based ranking refinement within a single mediation framework. Addressing this gap forms the basis of the proposed system presented in this work.

### III. METHODOLOGY

The method explains the construction and operationalizing of an integrated recruitment mediation system that oversight the entire hiring process via streamline and systematized evaluative phases. The system functions as a gate pipeline whereby participants go through a series of stages: resume gate keeping, an assessment test, an interview and an assessment ranking based on a set of pre-established criterions. Each phase yields outputs that are final and reasonable and these outputs are integrated to ensure the recruitment decisions are rational and consistent.

#### A. System Overview and Workflow

The proposed system is a recruitment mediation framework that automates all aspects of recruitment into a single system. The system acts as a middle-man between recruitment and candidates and manages job configuration, application, intake, evaluation, progression and final ranking within a sealed system.

The process starts with the recruiter creating a job posting along with defining the roles, required skills and experience criteria, as well as internal evaluation preferences. These parameters become the evaluation constraints for the entire process. After the job is published, candidates can view the job postings and submit their applications via a standardized intake system. Here, all applicants are treated the same and none ranked or filtered.

After the candidate submits their application, they continue to a multiple steps evaluation system with various phases. Each gate has its own evaluation function and its own rule for who can pass through. A candidate who does not meet the criteria for a specific gate is held back from moving forward. This way, candidates that pass through gates are ascending to the next steps, as this system guarantees aligned, transparent and incremental evaluation based on the defined criteria from the recruiter.

The system keeps ranking logic apart from eligibility enforcement. This is because eligibility is determined using a clear and rule-based explainable and reproducible system. Ranking is only done after all conditions have been met. This way, comparisons between unqualified candidates are avoided and greater consistency is achieved in the end results.

After the evaluation is done, the system creates a ranking for all the eligible candidates for every job posting. Recruiters look at these rankings and the score breakdowns for these candidates and they go on to shortlist and continue the hiring process in their external systems. The system then finishes its role as a mediator by informing the candidates and completing the recruitment cycle.

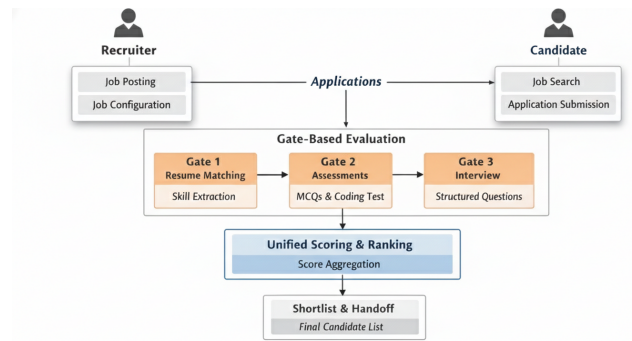


Fig. 1. System Architecture of the Recruitment Mediation Platform

#### B. Role-Based Access and Job Configuration

The recruiter starts the recruiting process by making a job posting and handling it through the various stages of the process. Job details include a description of the role, skills needed, level of experience, where the job is located and how much the job pays. Recruiters also mention their preferred internal assessments, which could range from technical assessments, specific score thresholds and weighted distributions, to the various assessment eligibility criteria. Candidates are not privy to these internal guidelines that control what happens in later stages of the assessment.

Candidates have their own portal where they can see job offers, apply for jobs and submit their resumes and any required tests. The application intake process allows candidates to apply to a job only once and makes sure that all applications are screened before they get to the evaluations. At this point, no ranks or filters are applied to the submissions, thus guaranteeing all candidates the same access as a starting point.

To avoid any cross-role and unauthorized access, the system handles both the authentication and authorization. Role access is governed at the API and workflow level, which means users are restricted from carrying out any activities associated with that role. For example, functions related to candidates, such as view or review candidates, are restricted to recruiters. This partitioning secures system information and improves system automated recruiting.

When the system collects all the applications for a job posting, it moves to the evaluation stage. This assessment takes into consideration the job-related parameters set by the recruiter focusing specifically on the screening, scoring and ranking steps. Such arrangement maintains an orderly and uniform flow from the point of application collection to evaluation by making criteria set by the recruiter control all the subsequent screening scoring and ranking processes.

### C. Resume Skill Extraction and Matching

The evaluation of candidates begins with reviewing the resumes to get key technical skills and qualifications and then match them to the requirements of the job posted by recruiters. Instead of simply matching keywords or recognizing tokens, the hybrid semantic resume evaluation system analyzes candidate skill sets using semantic embeddings and context, going beyond basic semantics and natural language processing.

When candidates submit resumes, the first document is processed by the resume extraction module, which transcribes the document to raw text. As the document is transcribed, several automated processes such as language validation, anonymization and text normalization are applied to the raw text. To establish trust in the operating effectiveness of the semantic models, the resume text is primarily validated in one language, English. To remove bias in the screening process, names, emails and phone numbers are made unknown to system. Additionally, the final text is normalized to remove anomalous formatting, excessive characters and spacing caused by the conversion of the documents.

After standardizing the text of the resume, the system goes on to perform sentence segmentation. This process breaks the text into smaller, more manageable pieces. These sentence segments serve as the foundational components for the system's algorithms to conduct semantic analysis relative to the expectations tied to the job opportunity.

To ascertain the presence of resumes that match the skills required by a particular job opportunity, the system utilizes a simplified transformer-based embedder model that translates text into numerical vectors. Each resume sentence is represented as a vector, as is each required skill that is mapped

in the job configuration. This collection of vectors enables the system to understand the meaning of the sentence beyond mere contextual relevance; the model comprehends the phrases that express the same meaning.

The relevance between a skill and the contents of a candidate's resume is determined by the skill's vector and the vectors of all resume sentences. The complete relevance analysis is performed through cosine similarity, a measure that evaluates how closely related the two vectors are in the semantic matrix.

The cosine similarity between two embedding vectors  $A$  and  $B$  is defined as:

$$Similarity(A, B) = \frac{A \cdot B}{\|A\| \times \|B\|} \quad (1)$$

where  $A$  represents the embedding vector of a required skill and  $B$  represents the embedding vector of a sentence extracted from the resume.

However, relying solely on semantic similarity may overlook additional signals that reflect the depth and relevance of a candidate's experience with a skill. The model seeks to remedy this by employing a number of corroborative signals that, when taken together, display the strength of a particular skill associated with a candidate's resume. These signals comprise semantic similarity, skill mention count, contextual relevance of the skill in the resume, and recency.

A strength score for each skill is calculated based on the weighted sum of the signals listed above:

$$SkillStrength = w_s S_1 + w_f S_2 + w_c S_3 + w_r S_4 \quad (2)$$

where  $S_1$  represents the semantic similarity score obtained from the embedding comparison,  $S_2$  represents the normalized frequency of the skill appearing in the resume,  $S_3$  represents the contextual importance of the skill based on its distribution within the resume text, and  $S_4$  represents the recency or temporal relevance of the skill if available. The parameters  $w_s$ ,  $w_f$ ,  $w_c$ , and  $w_r$  are configurable weighting factors that control the influence of each component in the overall skill strength calculation.

Once the system gets all the semantic scores, then it uses the final scoring formula to evaluate the scoring of the candidate. This scoring is used to check how much the candidate resume is matching the job requirements given by the recruiter.

The final skill score for a candidate is computed as the average strength across all required skills.

$$SkillScore = \frac{1}{N} \sum_{i=1}^N SkillStrength_i \quad (3)$$

where  $N$  represents the number of required skills specified for the job role and  $SkillStrength_i$  represents the strength score of certain skill in the candidate resume matching to job requirement

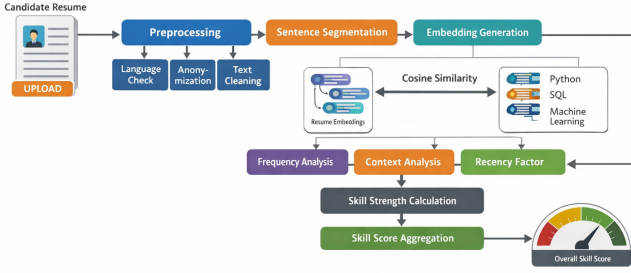


Fig. 2. Resume Semantic Skill Evaluation Pipeline

#### D. Assessment Generation and Evaluation

The assessment stage aims to measure the ability of candidates to handle skills relevant to a particular job after they have passed the first phase of screening resumes. There are two modes of assessment available. In cases where recruiters provide their own assessments, the platform functions as an execution and evaluation layer by securely delivering the assessment, collecting responses and generating scores. When recruiters do not upload a custom assessment, the platform generates an evaluation using an AI and NLP-assisted selection mechanism.

While building assessments generated by the platform, the company focuses on building assessments system's critique system's critique question bank from each technical evaluation question. The first question bank is built by sourcing questions from standard textbooks, interview prep materials, content from open-source repositories and faculty-reviewed content. All companies assess the accuracy, clarity and appropriateness of each question. Each question comes with metadata which includes the following: skill, domain, level of difficulty, type of question, length of answer and time to answer, which enables more flexible context-based personalization beyond simple skill taxonomy.

The system employs natural language processing to analyze job descriptions and identify the context for which particular competencies were needed. Rather than viewing competencies as static, the system reads the descriptions to identify the purpose and target domain of the role being referenced, including web development, backend development or data-related roles. Each assessment question is also represented within the system by semantic embeddings as a result of the text associated with the questions. As a result, the system is able to measure the semantic correlations between the job descriptions and questions to ensure that the identified questions reflect the intended purpose of the skill within the specific job role. Consequently, irrelevant questions, by default, lose priority without relying on strict exclusion criteria.

For candidates who pass through the previous evaluation gate, assessment questions are picked by ranking questions

based on their semantic relevance to the job description and matched resume signals. Mandatory skills impact selection priority while optional skills supplement the coverage. When coding assessments are set, coding questions are picked using the same relevance-based selection. Code submissions are assessed in a sandbox environment with specific execution and resource limits, while closed questions are graded against correct answer templates. Scores are consolidated and normalized to generate an assessment score that can be merged into subsequent evaluation stages.

The assessment framework is built to be iterative, based on the methodical refinement of the question bank. Performance indicators, such as completion rates, score distributions and question indicator anonymized etc. are analyzed to signal ambiguous and low-performing questions. Questions can be revised, replaced, or removed based on these indicators, while new questions are added with the same semantic indexing and validation process. This process will enable the system to improve the quality of assessments with no changes to the core logic of the evaluations and without introducing any risk of evaluative instability.



Fig. 3. AI/NLP-Based Assessment Selection and Evaluation Flow

Fig. 3 illustrates the AI and NLP-assisted assessment selection and evaluation process.

During assessments, the automated systems exercise the least possible amount of rule-based, consistency controls only to apply arbitrary limits on the length of assessments and to set minimum thresholds of eligibility for participants. These do not set boundaries on question appropriateness and operate as boundary controls for consistency and reliability. Recruiters are fully empowered to manage the assessment and may, at any point, disregard or shut off evaluations that the system generates. This framework allows the assessment to be flexible, scalable, contextually appropriate and ethically sound.

#### E. Structured Interview Evaluation

The structured interview stage is the last checkpoint before seller assessments and revolves around gauging fundamental communication skills, role relevant comprehension and the clarity of their answers. This stage is kept simple and more of a lightweight, more controlled than the rest of the cognitive load. This means no complex behavioral forecasting, sentiment analysis and emotional forecasting, but still provides recruiters a more valuable finish on the qualitative spectrum.

Assessment stage qualified sellers get pushed to the next step of this interview process: a brief, time bound interview

via the platform. This interview consists of a set of pre-defined questions that they all, across the board, get asked for that role. The questions aim to measure how well they explain themselves and how well they understand the role, as well as their ability to articulate a problem. Questions do not aim to measure any traits that pertain to that seller's personality.

There is uniformity to the interview process as well, with time limits and questions as well. Text or audio recording of their answers, predictive behavioral analysis and all the other ed tech buzz words are not utilized during this stage. The criteria bias and all the other ed tech buzz words are not utilized during this stage. The criteria and what is being measured is very clear.

Interviews are scored based on recruiting criteria, and initial scoring is treated as another data point, not as a determiner. Candidates that do not meet interview eligibility criteria are not progressed. Those that meet the criteria move on to aggregation.

Limiting the scope of interview evaluation to more closed criteria creates a balance between qualitative assessment and the consistency of this as a gate. The results at this stage provide to the candidate profile with minimal 'black box' decision making and subjectivity.

#### F. Unified Scoring and Candidate Ranking

After candidates complete all applicable evaluation stages, the system aggregates individual evaluation outputs into a unified scoring framework. This stage is responsible for combining signals obtained from resume screening, technical assessment, and structured interview evaluation to produce a comparable and explainable ranking of candidates for each job posting.

Each evaluation stage contributes a normalized score that reflects candidate performance within that stage. The skill alignment score is from resume screening, the assessment stage results in a technical score, and the interview stage assigns a score for qualitative communication. These are combined using a recruiter's custom weight distribution that reflects role-specific hiring priorities, and are not equal by default.

The overall score is a weighted average from each stage. These weight values can be set by the recruiter at job setup, and provide flexibility on which evaluative dimensions are more significant. For instance, the weighting can be balanced to reflect that technical roles are likely to assess higher on the assessment, while roles that are more communicative are likely to score higher on the interview. This approach is especially aligned to the outcome that is intended to be achieved by the recruiter.

To avoid any single score dominating the evaluation, the scores are adjusted for gaps and over thresholds. Only applicants that meet minimum eligibility requirements for all stages are evaluated, and only those candidates are included in the final ranking to ensure fairness in the outcome.

Ranked lists are accompanied with score breakdowns and transparency is given to the evaluators. This transparency helps

the evaluators see how each candidate ended up with their score which helps make better decisions. Recruiters are given the automated evaluation pipeline and scoring framework to complete candidate reviews and make selections.

$$\text{FinalScore} = w_1 \cdot rs + w_2 \cdot as + w_3 \cdot is \quad (4)$$

where:

- *rs* denotes the resume score obtained from Gate-1
- *as* denotes the assessment score obtained from Gate-2
- *is* denotes the interview score obtained from Gate-3
- $w_1, w_2, w_3$  are recruiter-defined weights such that  $w_1 + w_2 + w_3 = 1$

#### G. Recruiter Dashboard, Notifications, and Handoff

Once unified scoring and ranking are completed, the system provides recruiters with a dashboard that displays candidates ranked per job posting and offers brief summaries of scores from all evaluation phases. This dashboard allows recruiters to assess candidates without auto making decisions for them.

Based on the organization's policy, Recruiters can directly shortlist candidates from the ranked list. When the system finishes creating the shortlist, it sends out emails to candidates regarding the status of their application. Shortlisted candidates receive updates regarding onward progression to the next steps in the hiring process, while others receive the corresponding updates about not being selected.

Once all candidates have been contacted, the system completes its function in the recruitment process by closing the cycle and supplying the recruiters with information on the shortlisted candidates. At this point, no more automated evaluations are carried out, marking a clear line from system-assisted screening to human intervention in the hiring process.

#### H. Technology Stack

The platform is designed using modular web-based architecture utilizing Python and Django. The backend uses Django and Django REST Framework to facilitate role-based workflows, secure API communication, and handle requests in a scalable manner. Access control across recruiter and candidate portals is done using JSON Web Tokens for secure authentication and authorization.

User profiles, job postings, applications, and evaluations are stored and managed in the database layer (MySQL), which is integrated using Django's ORM. The responsive user interaction and seamless backend logic integration frontend was created using HTML, CSS, Bootstrap, and Django templates.

The components that utilize artificial intelligence and natural language processing are built with custom algorithms and minimal machine learning techniques. An example of skill extraction is done through custom-built logic for parsing and extraction, while scoring and ranking are done with a structured rule-based engine. Contextual and semantic relevance is processed using hybrid ML techniques (a combination of PyTorch, embedding-based models, and TF-IDF). This design focuses on transparency and contextual intelligence rather than large, opaque models.

Other supportive technologies include Python libraries for resume text extraction, Git for version control, and Postman for API testing. Email notifications are handled with basic SMTP. The architecture of the system aims to be extensible and allows for the upcoming use of Celery and Redis for background task processing, Docker for safe code execution and deployment, and Unicorn with Nginx for service provision at a production level.

#### IV. RESULTS AND EVALUATION

The evaluation was performed using a candidate resume containing the following technical skills: Python, Machine Learning, TensorFlow, HTML, CSS and JavaScript. These skills were extracted and analyzed by the semantic resume evaluation engine to determine their alignment with different job role requirements.

TABLE I  
RESUME EVALUATION RESULTS FOR DIFFERENT JOB APPLICATIONS

| Job Role                                                                                   | Score | Decision    |
|--------------------------------------------------------------------------------------------|-------|-------------|
| Full Stack Developer<br><i>Required Skills: Python, React, SQL, APIs</i>                   | 29%   | Rejected    |
| Machine Learning Engineer<br><i>Required Skills: Python, ML, TensorFlow, Data Analysis</i> | 81%   | Shortlisted |
| Frontend Developer Intern<br><i>Required Skills: HTML, CSS, JavaScript, UI Design</i>      | 61%   | Shortlisted |

Table I shows the ability of the semantic resume evaluation engine to assess candidate resumes based on varying job role criteria. The engine scans the resume and pulls out technical skills and matches them to the skills required in the job configuration. Candidates with higher evaluation scores and matches for the detected and required skills are shortlisted; candidates with less or no matches are more likely to be rejected. An example of this is with the role of Machine Learning Engineer, who scored with a match due to the inclusion of skills related to the required competencies of machine learning, while for the role of Full Stack Developer, this score was significantly lower due to the absence of several required technologies in the resume.

#### V. CONCLUSION

This paper described an innovative framework for the use of AI and NLP in the mediation of recruitment processes in order to enhance fairness, consistency, and transparency in practice for the initial stages of hiring. The system proposed incorporates automation: skill extraction from resumes, generation of assessments based on different contexts, evaluation of structured interviews, and consolidation of scores to assist the decision making of recruiters while retaining the human element from the system.

Providing the system employs NLP techniques that are light and coupled with an explainable metric for scoring, the system can avoid forming dependency with a black box system and also reduce the bias caused by score manual assessments. The system's design provides the capability for recruiters to

configure the system to their preference and also allows the system to improve (or enhance) incrementally without affecting the fundamental evaluation logic of the system. Overall, the proposed system exemplifies the appropriate, constructive, and scalable use of AI and NLP technologies in recruitment processes.

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